CMPE – 211

Preliminary Work (Pre-Lab Activity)

Laboratory Experiment # 5

Textbook Material: Chapter 9 *«Objects and Classes»*
Chapter 10 *«Object-Oriented Thinking»*

• • • TASK 1

Design a Java class named **Rectangle** to represent a rectangle. The class contains:

- ✓ Two **double** data fields named **width** and **height** that specify the width and height of the rectangle. The default values are **1** for both **width** and **height**.
- ✓ A no-arg constructor that creates a default rectangle.
- ✓ A constructor that creates a rectangle with the specified **width** and **height**.
- ✓ A method named **getArea()** that returns the area of this rectangle.
- ✓ A method named **getPerimeter()** that returns the perimeter.

Write a Java test program that creates two **Rectangle** objects—one with width **4** and height **40** and the other with width **3.5** and height **35.9**. Display the width, height, area, and perimeter of each rectangle in this order.

• • • TASK 2

Design a Java class that is named as **MyInteger**. The class should contain:

- ✓ An **int** data field named **value** that stores the **int** value represented by this object.
- ✓ A constructor that creates a **MyInteger** object for the specified **int** value.
- ✓ A getter method that returns the **int** value.
- ✓ The methods **isEven()**, **isOdd()**, and **isPrime()** that return **true** if the value in this object is even, odd, or prime, respectively.
- ✓ The static methods **isEven(int)**, **isOdd(int)**, and **isPrime(int)** that return **true** if the specified value is even, odd, or prime, respectively.
- ✓ The static methods **isEven(MyInteger)**, **isOdd(MyInteger)**, and **isPrime(MyInteger)** that return **true** if the specified value is even, odd, or prime, respectively.
- ✓ The methods **equals(int)** and **equals(MyInteger)** that return **true** if the value in this object is equal to the specified value.
- ✓ A static method **parseInt(char[])** that converts an array of numeric characters to an **int** value.
- ✓ A static method **parseInt(String)** that converts a string into an **int** value.

Write a Java test program that tests all methods in the class.

• • • TASK 3

The **String** class is provided in the Java library. Provide your own Java implementation for the following methods (name the new class **MyString**):

```
public MyString(char[] chars);
public char charAt(int index);
public int length();
public MyString substring(int begin, int end);
public MyString toLowerCase();
```

```java
public boolean equals(MyString s);
public static MyString valueOf(int i);
public int compare(String s);
public MyString substring(int begin);
public MyString toUpperCase();
```

Write a Java test program that tests all methods in the class.

● ● ● Sources

- Introduction to Java Programming and Data Structures, Comprehensive Version, 11th Edition, Y. D. Liang, Pearson, 2019.